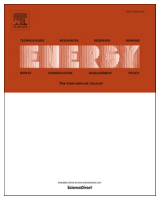
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Recovery and utilisation of waste heat for sponge iron process through combined preheating and power generation

Gajendra Kumar Gaurav [a,b], Shabina Khanam [b,*], Fuchun Zhang [a], Xinghui Liu [c,d,e,**]

[a] School of Physics and Electronic Information, Yan'an University, Yan'an-716000, China
[b] Department of Chemical Engineering, Indian Institute of Technology Roorkee, Roorkee - 247 667, India
[c] Department of Materials Physics, Saveetha School of Engineering, Saveetha Institute of Medical and Technical Sciences (SIMTS), Thandalam, Chennai, 602105, Tamilnadu, India
[d] Science and Technology on Aerospace Chemical Power Laboratory, Hubei Institute of Aerospace Chemotechnology, Xiangyang 441003, China
[e] Division of Research and Development, Lovely Professional University, Phagwara, India

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ABSTRACT

This paper deals with an option accounted for total waste gas heat recovered through combined preheating as well as power generation of feed material, kiln air, as well as slinger coal when it reaches 250 °C from 994 °C. The heat recovery is obtained through GCC Profile is plotted using HINT software. The power produced is 2.757 MW with an optimal condition having a temperature of 280 °C saturated, 600 °C superheated, as well as superheated pressure of 350 bar. The heat required around the superheater, boiler and preheater is 5066.9 kW, 12,669.93 kW and 6873.6 kW. The air requirement for this option is 52.588 t/h after iteration. Also, the iterative waste gas flow rate is 22.31 kg/s. Three new heat exchangers are required to achieve possible heat recovery by designing HEN through Pinch Analysis. The water requirement around EC is 14.88 t/h and 7.8 t/h as in the case of actual and modified HEN. The calculated pressure drops of different ducts having lengths 25 m and pressure 0.034 atm, 80 and 105 m are obtained pressure as 0.034, 0.12 and 0.16 atm. 3 FD fans are required for maintenance of required pressures in the duct. The TAC is INR 8068.8/y; payback period is 2.62 y.

1. Introduction

Sponge iron is manufactured in the form of solid by direct reduction of iron ore. Due to its honeycomb structure, it is called 'Sponge Iron'. The sponge iron quality is measured through % metallization, which is defined in the fraction of metallic iron i.e. Fe to the total iron available in the final product [1]. This is used in the steel manufacturing process as a predominant source. Agarwal and Sood [2] found that the required energy quantity of sponge iron for the kiln is available in the range of 14.63–20.9 GJ/t. A huge heat quantity is produced during the operation, which is not utilised in the system. So, the problems associated with energy are the major critical troublesome in this type of industry. Many investigators suggested numerous ways to the utilisation of waste gas heat in the process. Bandyopadhyay et al. [3] observed that bulk volatile matter (VM) quantity was lost with hot waste gases at the kiln feed end. Elsenheimer and Serbent [4] supported discussed feature and revealed that waste gas contained around 40 % of total kiln energy. They suggested numerous options to improve energy via the heat of waste gas, but they didn't discuss its practical difficulties as well as saved energy with the installation of these options. Hajidavalloo and Alagheband [5] incorporated preheating of raw materials using waste gas heat before admitting to the electric arc furnace. Sponge iron productivity was enhanced and energy consumption was reduced by 13 % and 14 %. Eriksson and Larsson [6] proposed different methods to increase process energy efficiency. These are available through the use of internal energy, the contribution of external energy and power generation. Mignard and Pritchard [7] discussed generation of marine energy in remote case and analyzed transmission and storage in the case of the sponge iron process. Biswas et al. [8] saved 10–12 % energy considering the reduced amount of air through axial as well as radial injection.

Kumar and Khanam [9] used waste gas heat for air preheating up to 170°C carried out using a plate heat exchanger. As a result, a reduction

* Corresponding author. Department of Chemical Engineering, Indian Institute of Technology Roorkee, Roorkee, 247 667, India.
** Corresponding author. Science and Technology on Aerospace Chemical Power Laboratory, Laboratory of Emergency Safety and Rescue Technology, Hubei Institute of Aerospace Chemotechnology, Xiangyang, 441003, China.
E-mail addresses: shabinahai@gmail.com, shabina.khanam@ch.iitr.ac.in (S. Khanam), liuxinghui119@gmail.com (X. Liu).

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in coal consumption was found up to 8.7 % correspondingly, waste gas was reduced up to 16.7 %. A water bath has been proposed by them for utilisation of available heat content around kiln discharge and found that 96.3 % water consumption was reduced in comparison to the existing system due to minimisation of heavier evaporation losses. Prasad et al. [10] suggested two cases case-1 and case-2 which were reckoned as firstly feed material preheating and secondly combined feed material and air preheating. They found that Case-2 was the most suited heat recovery alternative in comparison to case-1 as well as the existing recovery process, whereas reduction in coal, as well as water utilisation, was calculated as 30.5 % as well as 72.6 %. Prasad et al. [11] considered three practicable options for best design and energy conservation. They found that air and feed material preheating was most suitable for kiln utilising waste gas heat. Consumption of coal was reduced up to 12.5 % as compared to the existing available system.

Jena et al. [12] proposed a measurable evaluation of the process, presuming air requirement, loss of dust particles, waste gas composition, etc. The heat of waste gas is used for power generation suggested by authors and also achieved thermal efficiency was 46 % in the process where approx. 33 % kiln heat vanished accompanied by waste gas. How to get or compute to perform this matter is not described by them. Agarwal and Sood [13] proposed and installed a waste heat recovery boiler (WHRB) for generating process steam as well as tapping waste gas heat remains goes to waste. The total generated heat from coal is not used in reduction; only 35 % heat is used. A few difficulties faced by them in the installation areas:

(i) Generation of steam quality is not acceptable to turbine application in the start-up,
(ii) Formation of accretion in the kiln,
(iii) (iii) Not proper feedstock availability as well as shut down also mix up steam generation, and
(iv) (iv) an enormous amount of dust particles as 35×10^6 to 40×10^6 kg/m^3 found in kiln contained by waste gas. Ulrich and Tondo [2] described the feasibility of the plant comprising a WHRB and mentioned that plant production potentiality of sponge iron having 1000 t/d or beyond this is practicable, whereas a module plant having 500 t/d would be questionable. This feature was also favoured by Agarwal and Sood [13].

Numerous studies have been done in other plants using WHRB other than the direct reduced iron (DRI) process. Sui et al. [14] studied energy as well as exergy efficiencies of the cement production processes. Without considering WHRB in the process, the values for energy efficiencies for the preparation system of raw material, a rotary system as well as preparation system of pulverised coal, were obtained as 39.4 %, 50.2 %, as well as 10.8 %. With the installation of WHRB, the energy efficiencies values for the preparation system of raw material, a rotary system as well a preparation system of pulverised coal, were predicted as 45.8 %, 55.1 %, and 15.5 %. They found that the saved amount of energy is 30.75 kWh/t for waste heat power generation, which signified numerous economic as well as environmental benefits. They also suggested several energy-saving strategies for power generation using waste heat as a degree of resource, energy-saving effect as well as energy utilisation. Wang et al. [15] discussed three models based on thermodynamics in detail considering single, dual pressure as well as Energy Integration systems using Pinch Method principles for energy integration, power generation and heat recovery optimization utilising waste gas associated with the industry. A Total site hierarchical method related to energy integration is proposed for increment in power generation added with fuel-saving [16]. Integration based energy system was most profitable as compared to the other two strategies observed by them. In the first strategy, heat recovery was carried out the waste gas treatment, while in the second strategy, enhancing power generation was considered using optimised parameters, efficient equipment for heat recovery, and a steam cycle. The last strategy was Energy Integration between the existing industrial process and the power generation heat recovery system. They found that Total Site Efficiency was improved by 13.9 % as compared to the original or existing process due to optimization as well as Energy Integration in the third strategy while 6.7 % and 6.9 % were achieved in the case of the first and second strategies [17]. Nduagu et al., 2022 stated The comprehensive life cycle assessment of three different technological pathways for DRI production: rotary kiln DRI process, coal gasifier DRI process, and natural gas reformer DRI process are described well [18]. An inventory approach based on hybrid life cycle is employed, involves literature data, industry consultation and process modelling, to setup baseline process metrics, along with a sensitivity analysis to explain the variabilities and uncertainties impact. Energy input of 25 % and 8 % are reported through flue gases and shell losses respectively, whereas the free moisture vaporization consumes almost 15 % energy input. The rotary kiln furnace shows energy efficiencies of 66.4 %, in case of waste heat recovery. The results are used in optimizing operating conditions and provide thermal effects of mineralogical ore composition [19]. Green transformation has been outlined for rethinking innovation in case of iron and steel for advanced performance [20]. The recent challenges, technoeconomic analysis and vigorous development in case of steel industries are outlined [21].

The research gap is very much clear from the literature that preheating and power generation using the heat of waste gas have been performed by very few authors in the literature in the case of DRI production processes. In the present paper utilisation of available heat of waste gas used in preheating and power generation together in sponge iron process. This paper focuses on heat recovery through preheating and power generation and also detailed discussion about profitability analysis has been described.

2. Description of sponge iron process

The simplified process flow diagram (PFD) of an Indian sponge iron process, of 500 t/d production capacity, is shown in Fig. 1 along with industrial data. In this process, raw material admits to the kiln from the feed end and because of rotation as well as inclination inside the kiln feed moves to the discharge end. The product collects at the discharge end. On the other hand, as per process demand air supplies into the kiln in the form of primary as well as secondary air as indicated in Fig. 1. A non-oxidising environment is maintained, so that product temperature is cooled down to 110°C from 1,067°C. At 110°C, solid products come out from the rotary kiln and enter to rotary cooler [13]. A detailed description of the process is discussed by Gaurav and Khanam, 2017 [21].

3. Description of energy-saving strategy and corresponding necessary data

A strategy is proposed for saving energy, where preheating of raw materials as well as power generation are considered simultaneously, which is a novel option for energy conservation in the sponge iron process. In this case, preheating of kiln feed, kiln air and slinger coal is carried out using the heat of waste gas. The remaining heat of waste gas is used for power generation. For the recovery and utilisation of waste heat, data on the sponge iron process is collected. The stream data for this strategy is shown in Table 1 where preheating of feed materials, kiln air and slinger coal streams are done up to 200 °C, 600 °C and 200 °C. Waste gas is available from 994 °C to 250 °C. For preheating 9271.99 kW heat of waste gas is used and beyond that waste gas heat is recovered through power generation.

Along with stream data, a few more data are required such as specific heats of various streams and cost data of different materials, which are reported in Tables 1 and 2. For the installation of a duct to carry waste gas from one place to another in the process, the distance between different equipment is shown in Table 3.

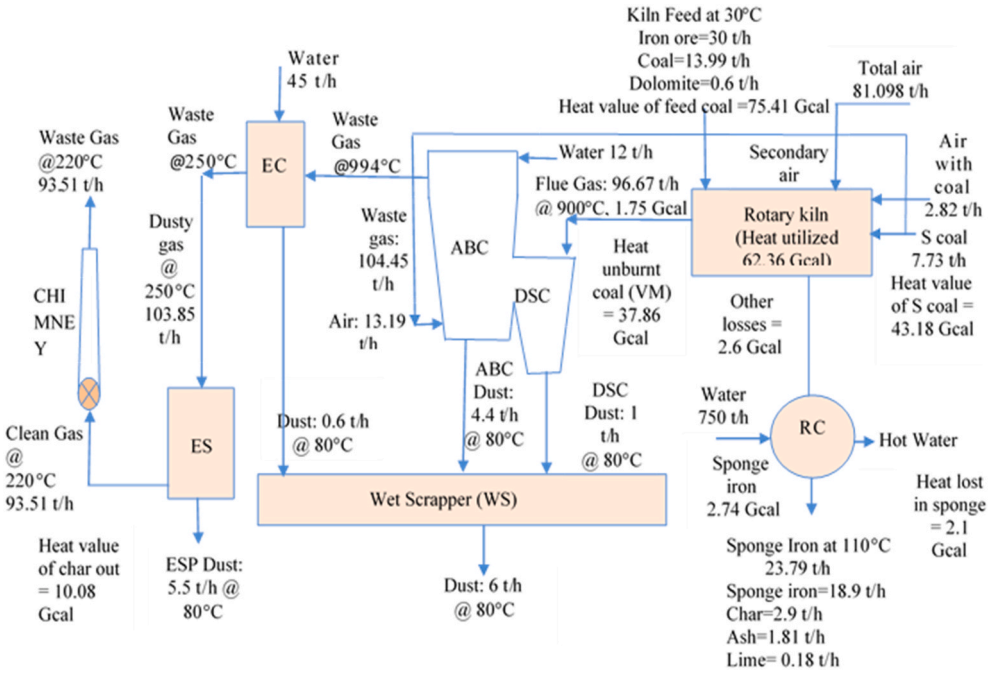


Fig. 1. Process Flow Diagram (PFD) of the sponge iron process.

Table 1
Data of different streams specific heat.

Stream	Specific heat (J/kg °C)
kiln feed	1092.7
Kiln air	1032.6
Slinger Coal	1884.30
Kiln outlet	687.9
Inlet to RC	4187
Waste gas	1140

Table 2
Cost operating data of different materials.

Material	Cost	
	Value	Unit
Iron ore	2600	INR/t
Coal (Mixing of feed coal and Slinger coal)	2500	INR/t
Standard Cost of Boiler Turbine System	46,800	INR/kW
Standard Maintenance cost	0.2	INR/kWh
The unit cost of power generation around the turbine	6.5	INR/kW
Fuel (HSD) for firing	33.4	INR/litre
Water	60	INR/kilolitre
Electricity cost	3.5	INR/kWh

Table 3
Temperatures profiles of waste gas in the duct.

For a length of 125 m		For a length of 80 m		For a length of 25 m	
Length, m	T, ^{0}C	Length, m	T, ^{0}C	Length, m	T, ^{0}C
0	600	0	906.2	0	994
10	592.8	10	905.6	10	993.5
20	592	20	905	20	993
40	592	40	904.3	25	992.7
50	591.5	50	904		
60	591.5	60	903.7		
70	591	70	902.8		
80	591	80	902.5		
100	589.6				
105	589.6				

4. Solution methodology

Different steps used to solve energy saving [22] strategy options are shown in Fig. 2. The expression to compute coal consumption for this strategy is detailed in the work of Gaurav and Khanam (Gaurav and Khanam, 2017) and shown in Eq. (1). Expressions involved in the estimation of total heat requirement for conventional process Hot utility is determined by pinch analysis [17]. Based on heat balance, it is found that heat utilised in the process is only 43.3 % of the total heat energy thus, term 0.433 is used. The, mass ratio of air to coal, practiced in actual plant is 3.8703. Now, air enters to the plant in three ways such as kiln air, in ABC and with slinger coal in mass fractions as 0.8504, 0.16107 and 0.03356, respectively. These are preheated up to different temperatures due to 600 °C, 200 °C, 200 °C respectively. Thus, different temperature approaches are applicable for these fractions such as $\triangle T_{a1}$, $\triangle T_{a2}$, and $\triangle T_{a3}$ are used for fractions 0.8504, 0.16107 and 0.03356, respectively. The average moisture contents are found in iron ore and coal are 2 % and 13 % by weight, respectively, in Indian conditions [23] and the amount of heat required to remove it. Considering all these values, fractions and temperature approaches and rearranging the resultant equation the final expression is found as:

$$\begin{aligned} Q_{hu}+Q_s+Q_p+Q_{loss}+(0.02\times M_s\times\lambda)=M_c[&NHV\times 0.433\\ &-\{3.8703\times((0.8504\\ &\times C_a(\Delta T_{a1}))+(0.03356\\ &\times C_a\times(\Delta T_{a2}))+(0.1611\\ &\times C_a\times(\Delta T_{a3}))\}\\ &-(0.13\times 2256.9)\\ &-(C_c\times(T_d-T_{sc})] \end{aligned} \quad (1)$$

where NHV = 22759.3 kJ/kg and $\lambda = 2256.9 kJ/kg$.

5. Results and discussion

The modification proposed through the present strategy considers preheating of feed material to a kiln, kiln air and slinger coal using waste gas heat. The remaining heat of waste gas is considered for power generation. Through this scheme total heat of waste gas is recovered

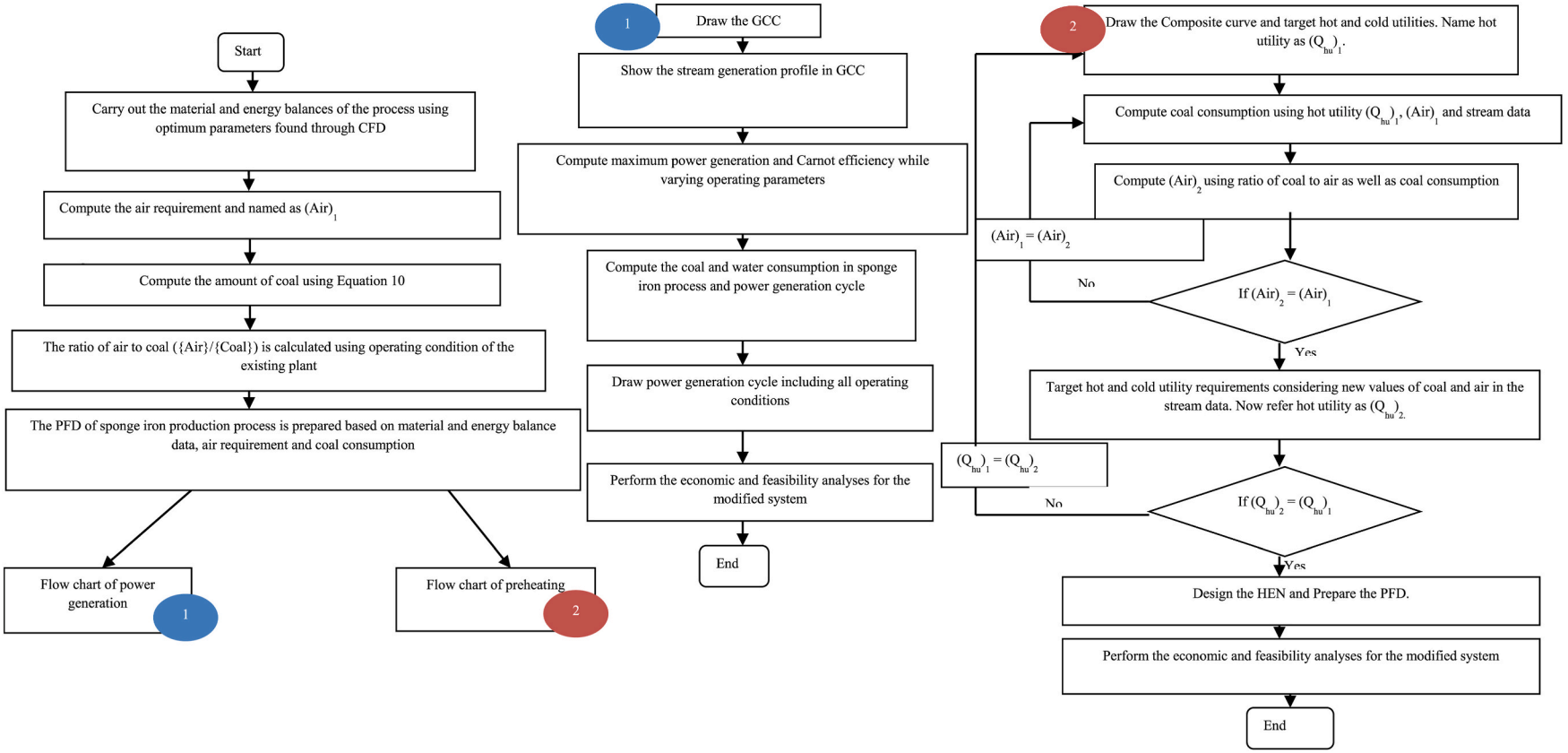


Fig. 2. Different steps used to solve energy saving strategy options.

during preheating and power generation, where it is cooled from 994 °C to 250 °C.

5.1. Coal and water consumption

For the present strategy, feed material, kiln air and slinger coal are preheated from 30 °C to 200 °C, from 30 °C to 600 °C and from 30 °C to 200 °C. For this purpose, 2271.83 kW and 10,782.6 kW and 503.74 kW of the heat of waste gas are required and consequently, it drops from 994 °C to 925.3 °C, from 925.3 °C to 599.4 °C and from 599.4 °C to 584.1 °C. It has been observed that waste gas should be cooled to 250 °C before entering ESP. Heat available in this from 584.1 °C to 250 °C is used for power generation. As consider ΔT_{min} as 50 °C and stream data shown in Table 4, the hot and cold utilities are computed as 0 and 11,052.3 kW, using Pinch Analysis. Due to preheating of the air, feed material and slinger coal up to 600 °C, 200 °C and 200 °C, in this strategy coal consumption, is reduced significantly as coal is utilised to heat air, iron ore and coal from 600 °C, 200 °C and 200 °C, to reaction temperature instead of 30 °C to reaction temperature.

To find the value of coal consumption using Eq. (1), a ratio of air to coal is maintained as 3.87, which is found considering flowrates of coal and total air utilised in the existing sponge iron [24] process. As preheating of kiln air, feed material and slinger coal are involved in this strategy, coal consumption for this scheme is computed based on the iterative method shown in Fig. 3.

It shows that the final value of coal consumption is found as 13.587 t/h, which is 37.4 % less compared the existing PFD. After preheating kiln air, feed material and slinger coal up to 600 °C, 200 °C and 200 °C, waste gas drops its temperature to 597.8 °C. Further, it is used for power generation to reach a temperature of 250 °C. For this purpose, the steam generation profile is plotted in the GCC as shown in Fig. 4.

It shows waste gas profile through solid line which is cool down from 572.8 °C to 225 °C considering ΔT_{min} as 50 °C. At the actual scale, it moves from 597.8 °C to 250 °C. The steam profile is also plotted at a shifted temperature in GCC. For power generation, it is assumed that saturated steam is produced at 280 °C whereas, superheated steam is generated at 350 °C and 70 bar. The steam profile in Fig. 4 shows that consuming 2604.78 kW of the heat of waste gas water is preheated from 80 °C to 280 °C. At shifted scale preheating of water is shown from 105 °C to 305 °C. At 305 °C it is drawn as a straight line as it is saturated steam, which consumes 4801.35 kW of the heat of waste gas. Finally, while taking 732.44 kW heat superheated steam is generated at 375 °C as shown in Fig. 4.

As a result, the temperature of waste gas decreases from 597.8 °C to 250 °C. Considering 85 % isentropic efficiency and 80 °C condensing temperature of superheated steam 2.28 MW of power is produced. As power generation depends on saturation and superheating conditions, it can be maximised while varying these parameters as shown in Fig. 5.

Fig. 5 shows overall power generation, Carnot efficiency and wetness fraction while varying the parameters like saturated temperature, superheated temperature as well as superheated pressure. The effect of variation of these parameters on power generation can be explained in as saturated temperature is varied from 200°C to 280°C considering fixed values of superheated temperature and superheated pressure as 350°C and 70 bar, respectively. The maximum power can be generated as 70 bar. However, to generate it value of wetness fraction is found as 0.155, which is almost equal to the maximum limit i.e. 0.15. Therefore, optimum saturated temperature is found at which maximum power generated. Further, with variation in saturated temperature from 200°C to 280°C, Carnot efficiency is fixed at 43 %, which is an obvious outcome as inlet and outlet temperatures of power generation cycle are fixed. Further, at optimum saturated temperature of 280°C and superheated pressure of 70 bar are considered constant with variation the superheated temperature from 300°C to 700°C. For such variation maximum power generation of 8.02 MW is found at 700°C superheated temperature as shown in Table 5. However, at these conditions' wetness fraction is found as negative and thus, it is neglected. Thus, the selected maximum power generation is 7.67 MW corresponding to 600°C, where wetness fraction and Carnot efficiency are predicted as 0.003 and 59 %, respectively.

Table 5 shows that the Carnot efficiency and power generation increase with increase in superheated temperature, whereas, wetness fraction decreases with increase in superheated temperature. Finally, superheated pressure is varied from 80 bar to 400 bar while fixing the values of saturated temperature and superheated temperature at optimum conditions i.e. 280°C and 600°C, respectively. For this case maximum power of 9.13 MW is generated with wetness fraction of 0.168 and Carnot efficiency as 57 %, which are corresponding to superheated pressure of 400 bar. As, wetness factor achieved at 400 bar is above than the tolerable value of 0.15. Therefore, optimum superheated pressure is accounted as 350 bar where power generation is 8.97 MW. Therefore, for generating 8.97 MW power the values of saturated temperature, superheated temperature and superheated pressure are considered as 280°C, 600°C and 350 bar, respectively as shown in Table 5. The final cycle of power generation with all operating parameters is shown in Fig. 6. Fig. 5 shows that a maximum of 2.82 MW power can be generated utilising waste gas heat. For generating 2.82 MW power, the values of saturated temperature, superheated temperature as well as superheated pressure are obtained as 280 °C, 548 °C as well as 250 bar. The total superheated steam generated is 2.779 kg/s. The power generation cycle is shown in Fig. 6 considering all parameters which are considered in operation in the cycle.

For this strategy, 10 t/h and 157.5 t/h of water are consumed in the power generation cycle as well as the condenser. In the modified system total coal and water consumptions are found as 13.587 t/h and 917.5 t/h. An amount of 13.587 t/h coal is utilised by HEN which is designed through Pinch Analysis [25,26] software-HINT considering ΔT_{min} as 50 °C as presented in Fig. 7.

For achieving possible heat recovery, a requirement for heat exchangers is needed of three in numbers. These two of them are (G-S) gas-solid, as well as one is G-G (gas-gas) exchanger. Along with these power generation systems are also employed. The effects of waste gas at various initial temperature such as 1020°C, 1035°C, 1050°C on output like coal consumption and power generation are shown in Table 5. It shows that it follows the same trends as discussed above in case of initial temperature 994°C. The water consumption will be increased in the similar manner because steam is increases with increment in temperature variation.

Table 4
Modified system stream data for Option-3.

Steam	Type	T_s	T_t	CP (kW/°C)	Unit
Waste gas	h1	994	250	23.402	ABC
Kiln air	c1	30	600	12.148	Rotary kiln
Kiln feed	c2	30	200	11.905	Rotary kiln
S coal	c3	30	200	1.903	Rotary kiln

5.2. Operability analysis

For the present strategy waste gas is used to preheat feed material, kiln air and slinger coal as well as to produce power. Before using the waste gas for these purposes, it should be cleaned as it consists of 6.1 t/h dust when it comes out from ABC. To remove these dust particles a high-temperature ceramic filter is installed after the ABC. it is noted that to preheat the feed material, kiln air and slinger coal waste gas moves from ABC to rotary kiln, which is in a backward direction than that maintained through an induced draft fan. Such movement of gas is possible using forced draft fans in the ducts.

Initially, waste gas moves from ABC to kiln inlet through an insulated duct of 25 m length to preheat the feed material from 30 °C to 200 °C, which is carried out using direct contact gas-solid heat exchanger of 2023.92 kW. Gas enters to 80 m duct to reach to discharge end of the kiln

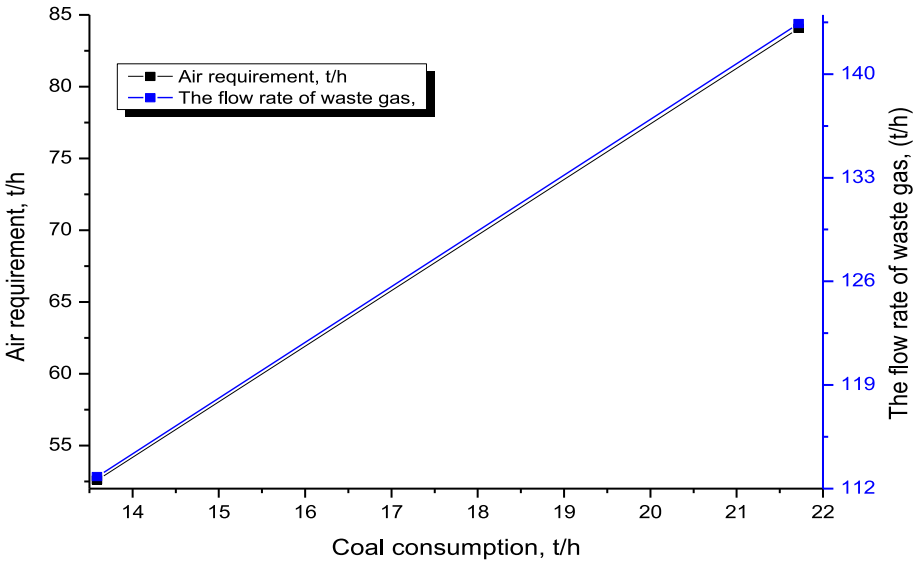


Fig. 3. Iterative Coal consumption, air requirement and flow rate of waste gas, t/h.

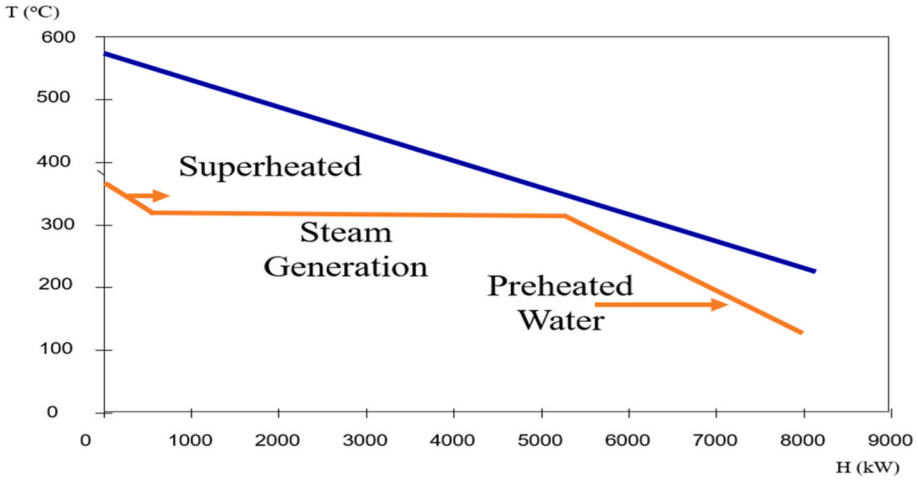


Fig. 4. Grand composite curve.

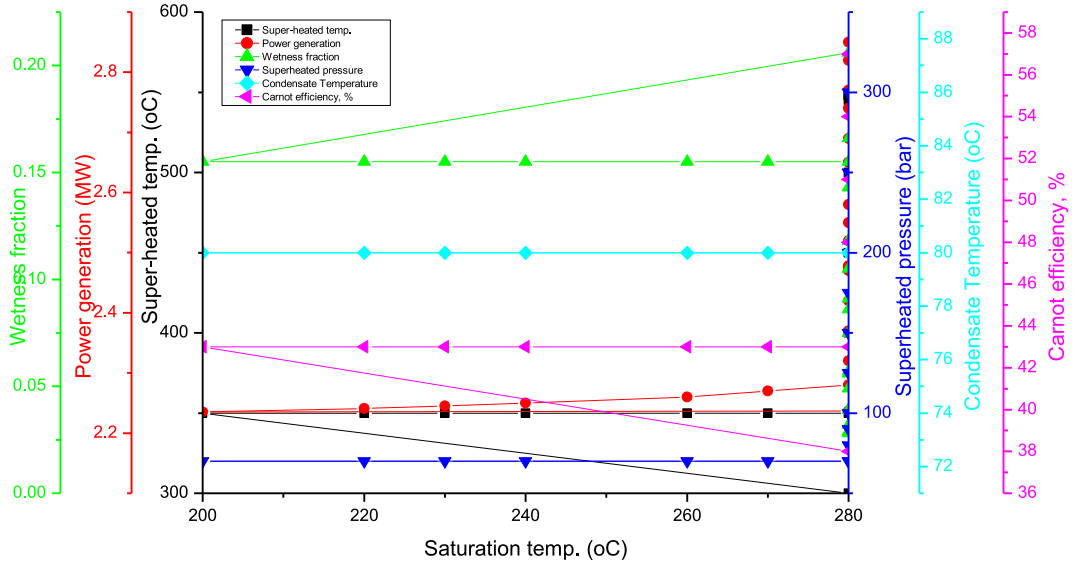


Fig. 5. Overall power generation, Carnot efficiency and wetness factor.

Table 5
Iterations results for calculation of coal consumption in Option-3.

S. No.	Saturation temp. (°C)	Super-heated temp. (°C)	Superheated pressure (bar)	Power generation (MW)	Wetness fraction	Carnot efficiency, %	Condensate Temperature (°C)
1	280	350	70	2.28	0.155	43	80
	270	350	70	2.27	0.155	43	80
	260	350	70	2.26	0.155	43	80
	240	350	70	2.25	0.155	43	80
	230	350	70	2.245	0.155	43	80
	220	350	70	2.241	0.155	43	80
	200	350	70	2.235	0.155	43	80
2	280	300	70	2.237	0.206	38	80
	280	400	70	2.32	0.118	48	80
	280	450	70	2.37	0.086	51	80
	280	500	70	2.42	0.056	54	80
	280	545	70	2.47	0.032	57	80
	280	548	70	2.476	0.029	57	80
	280	550	70	2.478	0.028	57	80
3	280	548	80	2.52	0.039	57	80
	280	548	90	2.55	0.049	57	80
	280	548	100	2.58	0.057	57	80
	280	548	125	2.65	0.075	57	80
	280	548	150	2.69	0.091	57	80
	280	548	175	2.74	0.105	57	80
	280	548	200	2.77	0.118	57	80
	280	548	250	2.82	0.143	57	80
	280	548	300	2.85	0.166	57	80

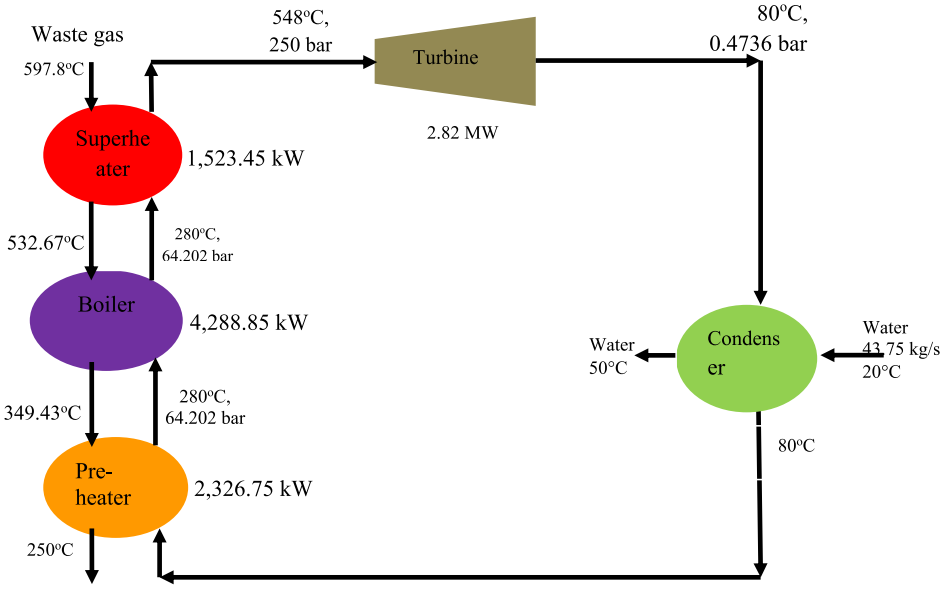


Fig. 6. Power generation cycle.

where it preheats kiln air from 30 °C to 200 °C using a shell and tube heat exchanger. Waste gas at 609.88 °C exchangers heat with slinger coal and preheats it from 30 °C to 200 °C using a gas-solid heat exchanger of 323.87 kW heat duty. Waste gas reduces its temperature to 597.8 °C as shown in Fig. 6. Waste gas at 597.8 °C enters to 105 m length insulated duct to move from kiln outlet to ABC outlet. At this temperature waste gas enters to power generation system where it supplies 2326.75 kW, 4288.85 kW and 1523.45 kW of heat in preheating, boiling and superheating, of water as evident from Fig. 4. For power generation 2.78 kg/s of water with CP 11.63 kW/°C is preheated from 80 °C to 280 °C using 2326.75 kW heat of waste gas. Water goes to a boiler where saturated vapour is formed at 280 °C and 64.202 bar using 4288.85 kW of the heat of waste gas. The saturated vapour enters to superheater where it is heated up to 548 °C and 250 bar pressure using 1523.45 kW heat. While supplying a total of 8139.04 kW of the heat of waste gas to the power generation system, it exits the system at 250 °C and then enters ESP as usually carried out in the existing system shown in Fig. 1. Lengths used for ducts are as per the distances between different equipment, shown in Table 3.

Therefore, in this strategy total of three ducts are required: first of 25 m length from ABC to kiln inlet, second of 80 m length from kiln inlet to kiln outlet and third of 105 m length from kiln outlet to ABC. All these ducts are insulated. The casing of these ducts is made of mild steel where outside insulation of glass wool is used. The temperature profiles [27] for 25 m, 80 m and 105 m long ducts are shown in Table 3. The exit temperatures of gas from 25 m, 80 m and 105 m long ducts are 992.7 °C, 902.5 °C and 589.6 °C as shown in Table 3. These are the actual temperatures of waste gas that affect the outlet temperatures of feed material, kiln air and slinger coal. Considering these temperatures of gas feed material, kiln air and slinger coal are preheated to 200 °C, 600 °C and 200 °C. Waste gas enters to power generation system at 589.6 °C and exits at 250 °C generates 2.76 MW of power instead of 2.82 MW. The

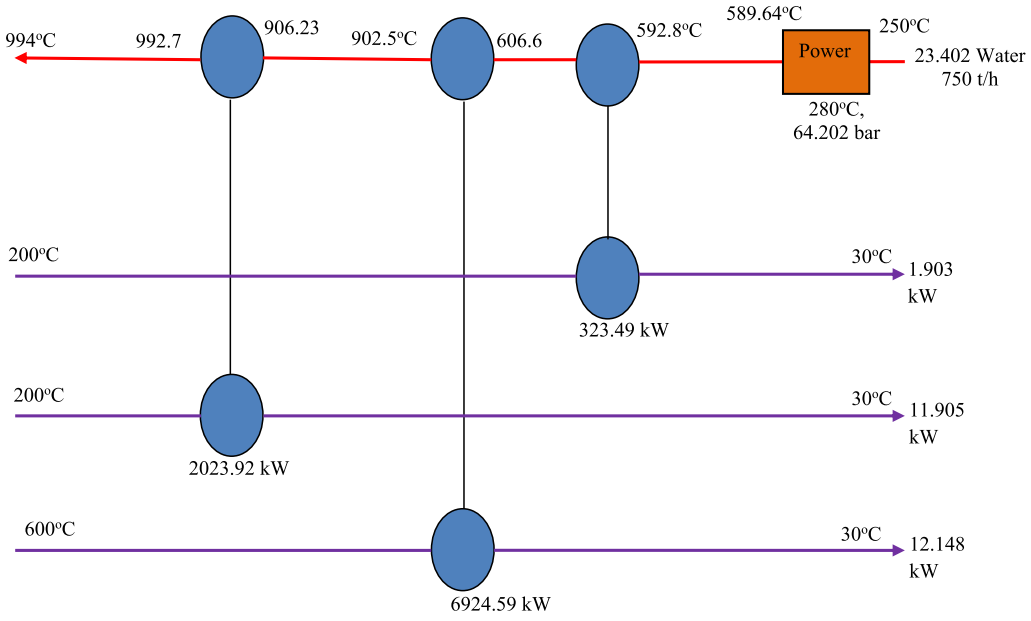


Fig. 7. Hen diagram of Option-3.

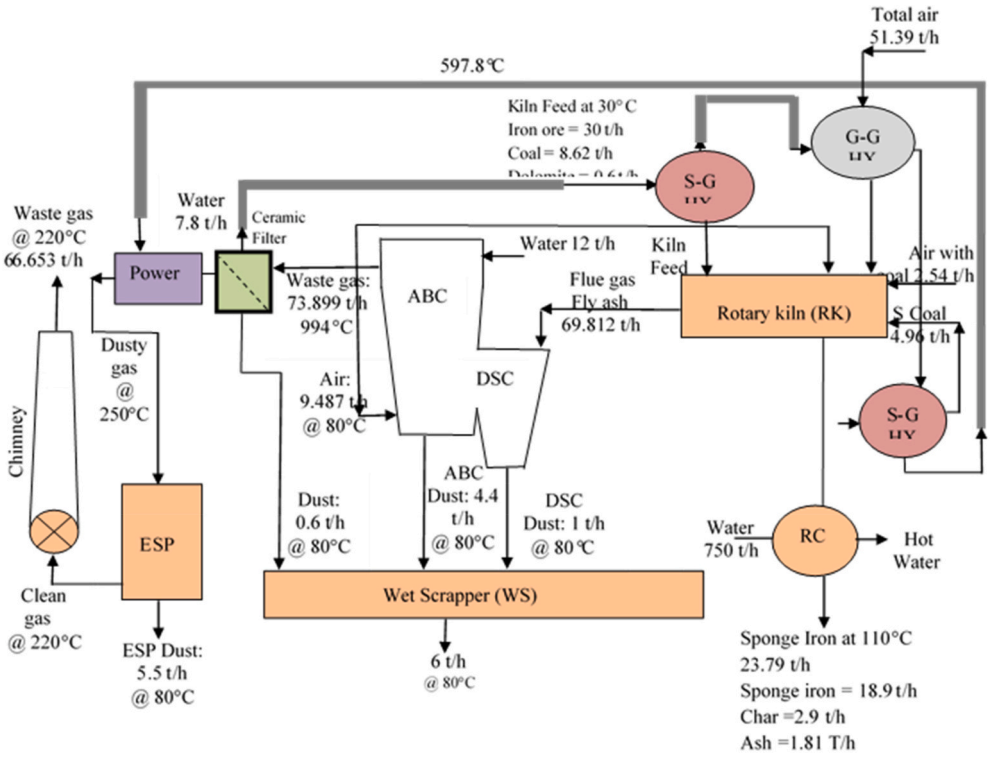


Fig. 8. Modified PFD of Option-3.

modified PFD of the present strategy is shown in Fig. 8.

There is a certain amount of pressure drop when the waste gas flows from ABC to the kiln and back to EC. The pressure drops for ducts are obtained using Eq. 2 and are found as 0.032 atm, 0.103 atm and 0.135 atm for 25 m, 80 m and 105 m long ducts. All these pressure drops have to be compensated through external means. For this purpose, 1 FD fan is used in the first duct of 25 m length to sustain 0.032 atm pressure. A duct of 80 m requires 1 FD fans for maintaining necessary pressures in the duct. 1 FD fans are required for a duct of 105 m length as air may move with a blower mounted at the periphery of a kiln. Thus, modified PFD for this strategy requires 3 FD fans. The pressure drop can be calculated using following equations: $\Delta P = \left[\left(0.109136 \times q^{1.9}\right) / {d_e}^{5.02}\right]$ ———— (2).

Where, ΔP = Pressure drop, (inches water gauge/100 ft of duct) q = Volumetric flow rate, (cfm - cubic feet per minute) d_e = equivalent duct diameter (inches).

5.3. Economic analysis

The economic analysis of the present strategy, which accounts for operating as well as capital costs, is discussed hereunder:

This strategy involves preheating of feed material, kiln air and slinger coal along with power generation. For this scheme operating cost is computed based on coal and water consumptions in the modified PFD, shown in Fig. 8. Based on 8000 h/y as working hours of the plant INR 2678.1 lakh/year and INR 4340.1 lakh/year are found for coal and water consumption. It is computed considering INR 2500/t and INR 60/kL for coal and water, as shown in Table 2. The operating cost of INR 577.5 lakh is predicted for maintenance of the filter based on the volume of dust gas handled by the ceramic filter per unit time. Due to regular tear and wear in parts of the boiler turbine system an expenditure of INR 44.112 lakh/year is invested in maintenance. Apart from this operating cost also includes the cost of electricity required to run 3 FD fans, which is predicted as INR 367.26 lakh/y as discussed by Gaurav and Khanam [25]. These costs for strategy are summarized in Table 7.

The scheme proposed through the present strategy involves a high-temperature ceramic filter, two gas-solid exchangers, one gas-gas exchanger, three ducts, 3 FD fans and a power generation system as additional equipment. The capital cost of a ceramic filter is computed as INR 2941.1 lakh [14]. Costs of gas-solid, as well as gas-gas exchangers, are found as INR 164.86 lakh and INR 50.8 lakh as discussed by Gaurav and Khanam [25]. The cost of the insulating duct of 25 m, 80 m and 105 m are computed as INR 9.628 lakh. As similar, the cost of the remaining two ducts is predicted as INR 95.21 lakh. The cost of 3 FD fans is computed and found as INR 367.26 lakh. The total capital investment for this strategy is presented in Table 6. Considering straight-line depreciation with 10 y as equipment life total annual cost (TAC) is found as INR 8086.8 lakh/year. A payback period of 2.62 years is predicted for this strategy as shown in Table 6.

The feasibility analysis of Option-3 is discussed as waste gas includes of an appreciable quantity of dust, which is deposited on tube surface in boiler, preheater as well as super-heater. The operation becomes impracticable/infeasible inside these because of regular deposition of dust. This issue can be controlled in two manners: firstly, periodically cleaning of the equipment should be happened for the removal of deposited dust and on the other hand, dust is drawn off first before admitting to the system of power generation. In initial case, regular washing/cleaning, as well as repairing/servicing of equipment, is required. Secondarily, clean gas is used at system entrance so that low equipment servicing is required. Second case is preferred here, as well as a ceramic filter associated with high-temperature is selected for this scheme in this work. It releases clean gas by trapping dust particles. Therefore, movement in waste gas is altered in case of modified PFD as compare to existing/actual PFD. Then, power generation system is being installed in path of waste gas. For ensuring feasibility of this movement, an induced type fan is connected afore chimney to ensure this movement is feasible. However, a large capacity induced type draft fan is needed to install in case of modified PFD as gas passes over heavier perplexed direction as compared to actual system.

A ceramic filter associated with high-temperature is installed subsequent to ABC as indicated in Fig. 9 used in modified PFD. This is a feasible key solution in dust removal of waste gas. FD fans are used in different ducts for making usual flow of waste gas movement in case of preheating of waste gas.

In the beginning, feed material is preheated via waste gas limit to 200 °C G-S HX which is of rotary type having direct connection with it, allocated nearby rotary kiln. Feed material enters to this exchanger at 30 °C via conveying belt. From feed end, waste gas entered to exchanger. Due to heat exchange, heated feed materials which is added to hopper directly, further goes to kiln, waste gas leaves exchanger at temperature of 992.72 °C.

After that, it moves via duct having length 25 m and reaches discharge end of kiln where gas admits to shell & tube type HX at 906.23 °C. The temperature reaches to 902.5°C as it passes through exchanger tubes. Accordingly, air is entered to kiln at 600 °C when it gains heat from 30.

°C to 600 °C. The hot air leaves shell & tube HX enters to chamber which is of hollow cylinder type directly which is located near to discharge end of kiln. The hot air permits to pass via nine different blowers, connected to nine ducts. On the other end, for maintaining a temperature of 600 °C to the kiln, a constant supply of hot air is allowed which is managed through these nine ducts which are directly connected with hollow cylinder. For maintaining airflow rate at high temperature i.e. 600 °C, metal dampers are used available in the blowers, has capacity to sustain high temperature of air. Air should not be entered to kiln at atmospheric temperature through nine blowers one should always ensure that hot air filled in the hollow cylinder. An assembly is suggested in which an adjustment of metal plate having circular shape is fitted inside hollow cylinder. This plate is attached with hot air outlet leaves the shell & tube HX. The condition should be static for circular plate as well as hot air pipe. Therefore, an assembly having U-shape is suggested which circulates along cylinder and inside assembly where plate is available. For sealing, this plate is used by avoiding mixed hot air in accompany with atmospheric air. At same rpm, assembly revolve successively with kiln as well as allow hot air entered to the kiln. Although, shell & tube HX, as well as duct in connection with it and plate, stand static so that it makes no attempt to shutdown kiln operation.

Preheating of slinger coal is done via G-S HX having direct contact type, where waste gas interchanges heat with air. The HX design, as well as functioning, is very much equivalent as applied in case of iron ore preheating. This is situated close by discharge end. Further, slinger coal allows near 30 °C through conveyer as well as warms till 200 °C. Waste gas enters into this exchanger at 606.6 °C and leaves at 592.8 °C. In pneumatic way, heated slinger coal leave via kiln discharge end. Moreover, waste gas passes through duct having length 105 m up to EC at a temperature of 432.1 °C and cooled down at a temperature of 250 °C utilising water having amount of 7.8 t/h.

6. Comparison of results of the present strategy with that of published work

The results of the present energy-saving strategy are compared with that of published studies based on coal as well as water consumptions,

Table 6
Economic analysis of Option-3.

Operating cost (Lakh/year)			Capital cost (Lakh)		Power (MW)	TAC (Lakh/y)	Profit (Lakh/y)	Payback period (y)
Commodity	Amount	Cost	Item	Cost				
Coal	13.587	2678.1	G-G HX	50.8	2.757	8086.8	1897.1	2.62
Water	917.5	4340.1	G-S HX	142.14				
Ceramic filter		577.5	G-S HX	22.72				
Electricity cost		74.5	Ceramic filter	2941.1				
The maintenance cost of the turbine		44.112	Ducts	9.628				
			Insulation	85.58				
			FD fans	367.26				
			Boiler & turbine system	1290.28				

Table 7
Comparison of different heat recovery options.

Option	Coal cons., t/h	Water cons, t/h	Operating cost, lakh/year	Electricity cost, lakh/year	Capital Investment, Lakh		TAC, lakh/year	Waste gas generated, t/h	Profit, lakh/year	Payback period, Years	Power Generation, MW
Existing System	21.72	795	3866.9	–	765		4431.3	128.376	728	1.05	–
1	21.72	1242.7	10308.9	–	Item Boiler and turbine system Ceramic filter	Cost 4197.96 4157.3	12104.3	98.358	2029.9	4.17	8.97
2	13.587	757.8	6262.8	74.5	G-G HX G-S HX G-S HX Ceramic filter Ducts Insulation FD fan	50.8 142.14 22.72 2941.1 8.97 42.79 367.26	7272.4	82.679	1126.8	3.22	–
3	13.587	917.5	7018.8	74.5	G-G HX G-S HX G-S HX Ceramic filter Ducts Insulation FD fans Boiler and turbine system	50.8 142.14 22.72 2941.1 9.628 85.58 367.26 1290.28	8086.8	82.679	1897.1	2.62	2.76

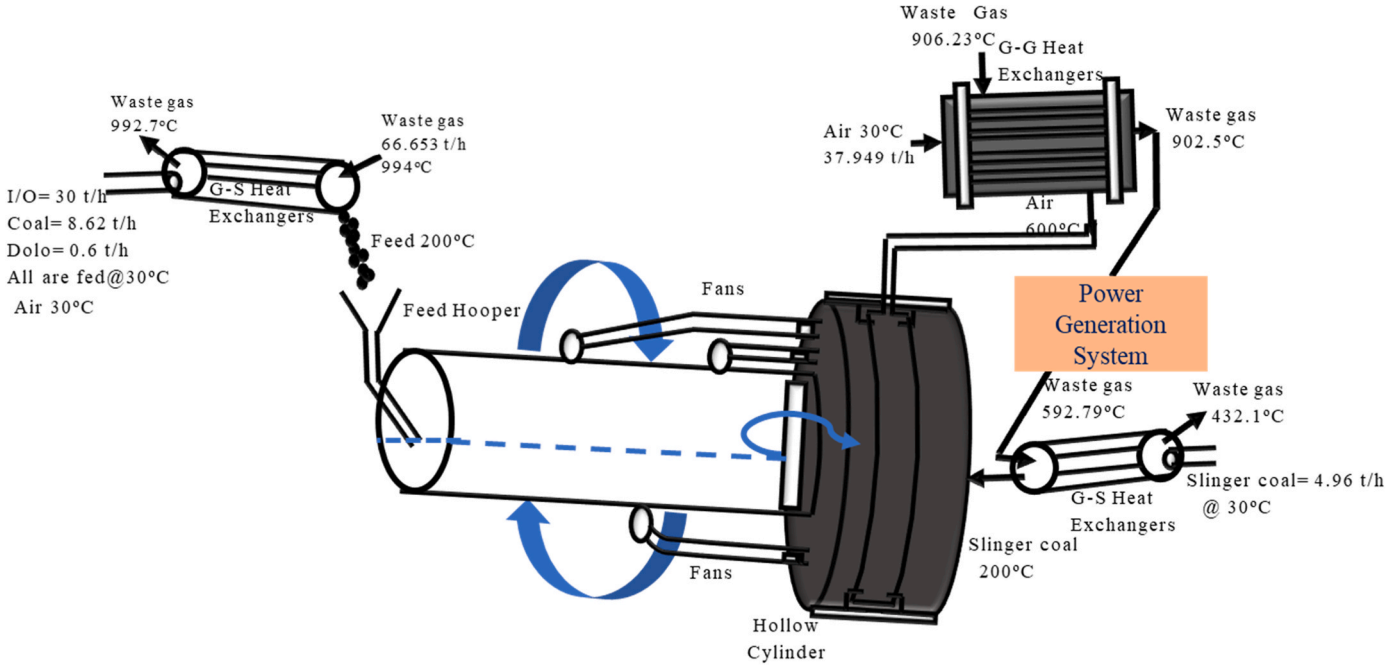


Fig. 9. Feasibility Diagram for option-3.

payback period, total required capital, profit, etc. and reported in Table 8.

Eight cases were proposed by Prasad et al. [10] for energy conservation. As compared to the existing system the best-case of Prasad et al. [10] has been observed to consume 93.7 % and 12.3 % less water and coal. In this case, three S-G and one G-G heat exchangers with an investment of INR 621.06 lakh are required. In the present strategy, a similar case study is chosen for energy conservation along with power generation. When compared with the best case of Prasad et al. [10], the present strategy has been found to have preheating of air up to 600 °C in instead of 300 °C by Prasad et al. [10]. The present strategy recovers more energy than that of the best case of Prasad et al. [10] due to the higher temperature of preheated air. As shown in Table 8, the reduction of coal in this strategy and Prasad et al. [10] have been found to be 37.4 % and 12.3 %. Coal saving in the present strategy is more compared to Prasad et al. [10], whereas compared to this strategy, the payback period for Prasad et al. [10] is very less. The reason for this observation is that in Prasad et al. [11], the percentage of water saving is 93.7 %

Table 8
Comparison of results of Option-3 with the published work.

S. No.	Strategy Name/Author name	Integration done in best case	Capital required, lakh INR	Coal saved %	Water saved %	Payback Period, years	Profit, lakh INR/year	Net Profit in plant life, lakh INR
1	Prasad et al. [11]	Preheating of air up to 300 °C with ESP exit	621.06	12.3	93.7	0.15	3092.6	30,457
2	Prasad et al. [28]	Preheating of feed material and air up to 110 °C and 385.7 °C	588.2	30.5	72.6	0.56	1180.2	11,141
3	Kumar and Khanam [9]	Air is preheated to 80 °C by kiln outlet using water bath	122.98	4.8	96.3	0.93	132.9	1206
4	Option-3 with preheating and power generation	Air preheated to 600 °C by kiln outlet. Feed and slinger coal preheated to 200 °C by ESP exit.	4909.51	37.4	37.2	2.62	1897.1	14,000

whereas 13.35 % more water is needed in this strategy, because of which there have been a many fold increase in profit for Prasad et al. [10] when compared to Option-3 as shown in Table 8. Feasibility of the modification proposed by Prasad et al. [10] has not been discussed while reducing 93.7 % of water. Waste gas has been proposed by them as a coolant instead of water in rotary cooler. Use of waste gas as coolant is not feasible in a rotary cooler as the cooler is a rotating device whereas waste gas comes from a stationary source such as ESP. On one hand the practical aspect of the best case of Prasad et al. [10] is not clear. On the other hand, in the present strategy water is consumed in the power generation cycle as well as condenser, which is a more feasible solution because here water is further used as steam. A ceramic filter [29] is also used as additional equipment along with different heat exchangers as well as FD fans. The cost of a ceramic filter is significantly higher in comparison to other equipment cost as shown in Table 7. Due to higher capital investment and lesser profit than that of Prasad et al. [11] payback periods are higher for this strategy.

A heat recovery option is evaluated when it is multiplied by total profit with the difference between plant life and the payback period. The prediction for the payback periods of the strategy are based on total profit and capital investment of the modification. The plant life considered in the present work and Prasad et al. [10] is 10 y. Due to the very less payback period for Prasad et al. [10] the net profit is significantly higher than that of the present strategy.

Two modifications were proposed by Prasad et al. [11] for utilising the heat obtained from waste gas. The existing system gets compared with these cases based on coal and water consumption. Other cases being capital cost, profit, and payback period. The best modification is obtained for the Case-2 where air and feed material are preheated up to 385.7 °C and 110°C. When the results of Case-2 are compared with of this strategy of present study, it comes out to be the combination of preheating of slinger coal, air and feed material up to 200°C, 600°C and 200°C. For this strategy, the preheating saving for coal is found as 37.4 %. However, for the Case-2, it is only 30.5 % as shown in Table 8. When compared to Prasad et al. [11], the payback period is significantly high for this strategy. This is due to fact that present strategy involves 8.35 times higher capital investment as compared to that of Case-2 of Prasad et al. [11]. Water consumption is 72.6 % less for Case-2 whereas there is an increment of 13.35 % in this strategy. Due to this, there is an increase in the profit for Case-2. For the present strategy, there is more payback periods because of more capital investment and less profit. The main drawback of the study being the absence of discussion by another on practical implication of modification in Case-2. Based on the plant life of 10 years, the net profit of the plant life is less than that of the present strategy.

Two cases were also proposed by Kumar and Khanam [9] for the recovery and utilisation of the heat of waste gas. 4.8 % less coal is consumed in the best case of Kumar and Khanam [9] which is 37.4 % in this strategy. This is because of preheating of air is done up to 600°C, which is significantly higher than the best case of Kumar and Khanam [9], where is preheating of air only up to 80°C as shown in Table 7. Payback period is very high for this strategy than that of Kumar and Khanam [9], this is again due to the similar facts as given by Prasad et al. [11]. From Tables 8 and it is evident that the profit for the present strategy is significantly higher than that of Kumar and Khanam [9] as huge amount of coal consumption is saved by this strategy as compared to the best case of Kumar and Khanam [9]. From Tables 8 and it is also evident that net profit in the plant life is higher for the present strategy due to more profit as compared to Kumar and Khanam [9] as shown in Table 8.

It is evident on the basis of above discussion that the result of the present work have been compared against the published literature. When compared with few studies such as Prasad et al. [11] and Prasad et al. [10], the present strategy shows lower profit than Prasad et al. [10] and higher profit than Prasad et al. [11].

7. Conclusions

(i) The proposed option of Heat Integration in this paper is reckoned with combined preheating as well as power generation which accounts for the total waste gas obtainable in the process. Option-3 uses total waste gas heat collected from the process, where it loses heat from 994 °C to 250 °C. The heat exits ABC has been utilised for Energy Integration. The prominent conclusions are: On the basis of the payback period combined preheating and power generation are considered the most cost-effective case as compared to preheating as well as power generation separately. In terms of net profits, Option-3 is more cost-effective compared to the other two options.

(ii) Option-3 is specified as a superlative option in accordance with heat integration. A valuable amount of coal consumption 8.13 % is saved which is higher with reference to Kumar work Kumar and Khanam [9]. Also, an amount of 37.2 % water is saved via integration in case of water consumption.

(iii) The reduction in waste gas is observed and reached 112.8 t/h from 143.4 t/h. This is a supplementary profit in the case of Option - 3.

(iv) The existing system utilises 7.45 times higher energy as compared to the theoretical value. It decreased to 4.69 times due to appropriate energy integration in the case of Option – 3, it saved high energy with reference to Prasad work saved 6.2 times only Prasad et al. [10].

(v) An amount of 2.757 MW of power is generated using waste gas heat present in the existing system from 994 °C to 250 °C. The operation of the power generation system is enhanced through proper solving of problems associated with scaling as well as the removal of dust that is deposited in boiler tubes and exchangers too.

CRediT authorship contribution statement

Gajendra Kumar Gaurav: Conceptualization, Investigation, Methodology, Validation, Writing – original draft. **Shabina Khanam:** Conceptualization, Supervision, Validation, Writing – original draft,

Writing – review & editing. **Fuchun Zhang:** Supervision, Writing – review & editing. **Xinghui Liu:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

[1] Chatterjee A. Sponge iron production by direct reduction of iron oxide. PHI learning Pvt. Ltd.; 2010. p. 82–127.
[2] Ulrich H, Tandon JK. The CODIR process for India–an example for optimum coal usage and its potential for energy recovery, vol. 2. Jamshedpur, India: Proceedings of Conference based on Alternative Routes on Iron and Steel; 1988. p. 21–5.
[3] Bandyopadhyay A, Ray AK, Srivastava MP, Subba Rao SVB, Prasad KK, Bandyopadhyay PK, Haque R, Choudhary BR. Selection of coals for rotary kiln sponge iron plant. Trans Indian Inst Met 1987;40:209–18.
[4] Elsenheimer G, Serbent H. The current position of the SL/RN process, taking into account conditions in India. In: Proceedings of international conference on alternative routes to iron and steel. vol. 2; 1988. p. 105–10. Jamshedpur, India.
[5] Hajidavalloo E, Alagheband A. Thermal analysis of sponge iron preheating using waste energy of EAF. J Mater Process Technol 2008;208:336–41.
[6] Eriksson K, Larsson M. Energy survey of the sponge iron process. Department of Chemical Engineering, Lund Institute of Technology 2004:1–4. Lund, Sweden.
[7] Mignard D, Pritchard C. A review of the sponge iron process for the storage and transmission of remotely generated marine energy. Int J Hydrogen Energy 2007; 32:5039–49.
[8] Biswas DK, Asthana SR, Rau VG. Some studies on energy savings in sponge iron plants. Trans ASME 2003;125:228–37.
[9] Kumar V, Khanam S. Recovery and Utilization of waste heat in coal-based sponge iron Process. Chem Eng Process 2012;56:19–28.
[10] Prasad AK, Prasad RK, Khanam S. Design modifications for energy conservation of sponge iron plants. J Therm Sci Eng Appl 2011;3. 015001-1.
[11] Prasad AK, Prasad RK, Khanam S. An investigation for generation of energy conservation measures for sponge iron plants using process integration principles. International Journal of Research and Reviews in Applied Sciences, IJRRAS 2011; 6:77–88.
[12] Jena SC, Patnaik NK, Sarangi A. Heat and mass balance in rotary kiln sponge iron making. In: Proceedings of international conference on alternative routes to iron and steel; 1996. p. 59–64. Jamshedpur, India.
[13] Agarwal VP, Sood KC. Direct reduction through coal route & power generation from the kiln waste gases. In: Proceedings of international conference on alternative routes to iron and steel; 1996. p. 51–8. Jamshedpur, India.
[14] Sui X, Zhang Y, Shao S, Zhang S. Exergetic life cycle assessment of cement production process with waste heat power generation. Energy conservation and management 2014;88:684–92.
[15] Wang Y, Niu Y, Zhang X, Wang Z, Wang S, Hui S. Optimization and energy integration of heat recovery and power generation system. Appl Therm Eng 2016; 107:294–300.
[16] Linnhoff B, Townsend DW, Boland D, Hewitt GF, Thomas BEA, Guy AR, et al. A user guide on Process integration for efficient use of energy. first ed. Rugby, UK: Institution of Chemical Engineers; 1982.
[17] Kemp IC, Analysis Pinch Integration Process. A user guide on Process integration for efficient use of energy. second ed. Butterworth-Heinemann, Elsevier Pvt. Ltd; 2007. p. 53–5.
[18] Nduagu EI, Yadav D, Bhardwaj N, Elango S, Biswas T, Banerjee R, Rajagopalan S. Comparative life cycle assessment of natural gas and coal-based directly reduced iron (DRI) production: a case study for India. J Clean Prod 2022;347:131196.
[19] Quintero-Coronel DA, Guillin-Estrada WD, Echeverri-Roman JL, Maury H, Corredor L, Ruiz JA, Rueda BS, Gonzalez-Quiroga A. Large- and Particle-Scale energy assessment of reduction roasting of nickel laterite ore for Ferronickel production via the rotary Kiln-Electric furnace process. Therm Sci Eng Prog 2022; 32:101331.
[20] Mallett A, Pal P. Green transformation in the iron and steel industry in India: rethinking patterns of innovation. Energy Strategy Rev 2022;44:100968.
[21] Shahabuddin M, Brooks G, Rhamdhani MA. Decarbonisation and hydrogen integration of steel industries: recent development, challenges and technoeconomic analysis. J Clean Prod 2023;395:136391.
[22] Choudhary Rajesh, Mukhija Abhishek, Sharma Subhash, Choudhary Rohitash, Chand Ami, Dewangan Ashok K, Kumar Gaurav Gajendra, Jaromír Klemeš Jiří. Energy-saving COVID–19 biomedical plastic waste treatment using the thermal - Catalytic pyrolysis. Energy 2023;264:126096.
[23] Agarwal BB, Prasad KK, Sarkar SB, Ray HS. Cold bonded ore-coal composite pellets for sponge iron making. Part 1 Laboratory scale development. Ironmak Steelmak 2000;27:421–5.
[24] Gajendra Kumar Gaurav, Khanam Shabina. Computational fluid dynamics analysis of sponge iron rotary kiln, Case Studies in Thermal Engineering 2017;9:14–27.
[25] Gaurav GK, Khanam S. Profitability analysis of power generation using waste heat of sponge iron process, vol. 141; 2017. p. 333–47.
[26] Prasad AK. An application of pinch technology for process integration in coal based sponge iron plant [PhD thesis]. India: National Institute of Technology Jamshedpur, University to be Deemed; 2011. p. 321–33.
[27] Gajendra Kumar Gaurav, Khanam Shabina. Analysis of temperature profile and % metallization in rotary kiln of sponge iron process through CFD. Journal of the Taiwan Institute of Chemical Engineers 2016;63:473–81.
[28] Prasad AK, Prasad RK, Khanam S. Development of energy conservations scenarios for sponge iron industry using process integration. Energy Efficiency 2011;4: 321–33.
[29] EPA-600/2-77-056 Evaluation of ceramic filters for high-temperature/high-pressure fine particulate control. 1977.

Nomenclature

C: specific heat, J/kg∘C
D: diameter of kiln, m
G: gas
h: heat transfer coefficient, kJ/h m^2
L: length of kiln, m
m: mass flow rate, kg/h
Q: heat, kW
S: solid
T: temperature
NHV: net heating value, kJ/kg
CC: capital cost, Rupees

Subscript

a: air
c: coal
m: moisture
p: process
h_u: hot utility
s: ore, supply
t: target
r: radiation
loss: from kiln
i: inlet to kiln

Greek

Lemda l =: latent heat of vaporization, kJ/k
INR =: Indian Rupee